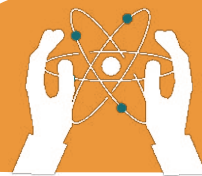


UNIT 5

ELEMENTS, COMPOUNDS, AND MIXTURES

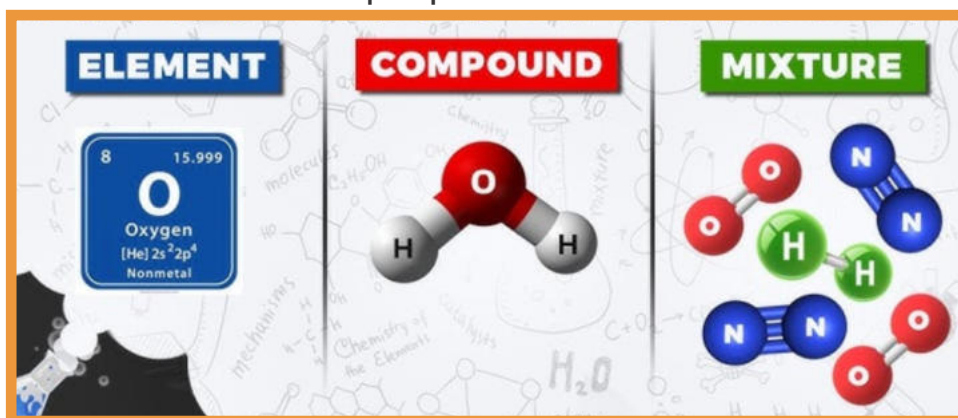
Learning objectives:

- Understand the differences between elements, compounds, and mixtures, and their key characteristics.
- Explain methods of separating mixtures, such as filtration, evaporation, magnetic separation and distillation
- Identify and classify substances as elements, compounds, or mixtures based on their properties



What are elements?

An **element** is a pure substance made up of only one kind of atom, while a **compound** is a substance made from two or more elements that are chemically bonded. A **mixture**, on the other hand, is a physical combination of two or more substances, where each retains its individual properties.



Key Concepts

Elements

Pure substances with only one type of atom.

Compounds

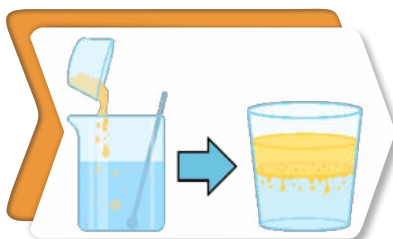
Substances with two or more different types of atoms bonded together.

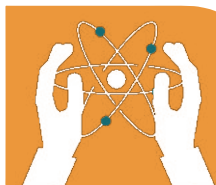
Mixtures

Physical combinations of two or more substances, not chemically bonded.

1. Elements

An element is the simplest form of matter that cannot be broken down by chemical means.





Characteristics:

1. Made up of One Type of Atom

- An element consists of identical atoms.

Example: Oxygen (O) is made up of only oxygen atoms, and Iron (Fe) is made up of only iron atoms.

2. Cannot Be Broken Down

- Elements cannot be split into simpler substances by chemical or physical means.
- **Example:** You cannot break gold (Au) into anything simpler—it is just gold.

3. Found in Three States of Matter

- **Solids:** Iron (Fe), Gold (Au), Silver (Ag)
- **Liquids:** Mercury (Hg), Bromine (Br)
- **Gases:** Oxygen (O), Nitrogen (N), Hydrogen (H)

4. Can Be Metals, Non-Metals, or Metalloids

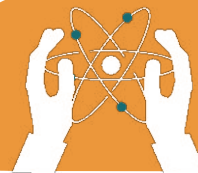
- **Metals:** They are good conductors of electricity and heat, shiny, malleable (e.g., Copper, Aluminium).
- **Non-Metals:** They are poor conductors, brittle, and dull (e.g., Oxygen, Carbon, Sulfur).
- **Metalloids:** They have properties of both metals and non-metals (e.g., Silicon, Boron).

5. Have a Fixed Melting and Boiling Point

- Each element melts and boils at a specific temperature.
- **Example:** Water boils at 100°C, but iron melts at 1538°C.

6. Each Element Has a Symbol

- Every element has a unique chemical symbol, sometimes derived from its Latin name.
- **Examples:**
 1. Hydrogen (H)
 2. Oxygen (O)
 3. Gold (Au) – from Latin "Aurum"

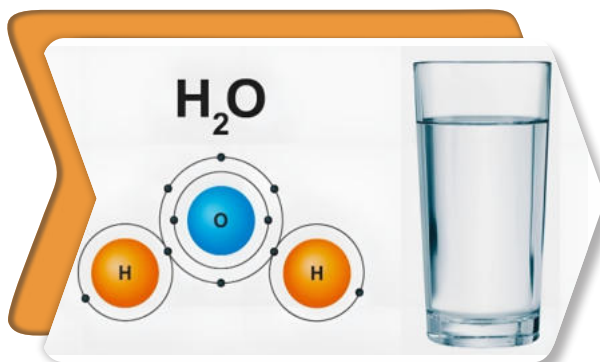


7. Can Combine to Form Compounds

- Elements react with each other to form compounds.
- **Example:** Hydrogen (H) + Oxygen (O) = Water (H₂O)

2. Compounds

A compound is a substance formed when two or more elements are chemically bonded in a fixed ratio.



Characteristics: Compounds have different properties from the elements that form them.

Examples:

Water (H₂O): It is made from hydrogen and oxygen.

Carbon Dioxide (CO₂): It is made from carbon and oxygen.

Inquiry based Activity

Hypothesis: "Compounds have properties different from their elements."

Instructions:

Combine sodium and chlorine to form sodium chloride (table salt), and observe its properties compared to those of sodium and chlorine.

Conclusion:

Sodium chloride's properties differ significantly from sodium and chlorine, showing that compounds are distinct from their elements.

DID YOU KNOW?

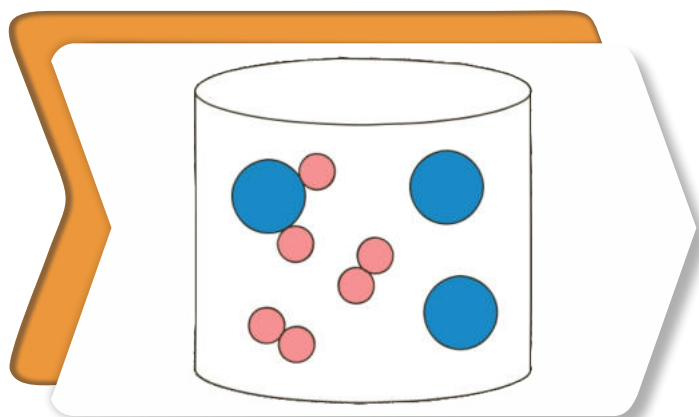
Elements can't be broken down chemically, but compounds can be separated into their elements





3. Mixtures

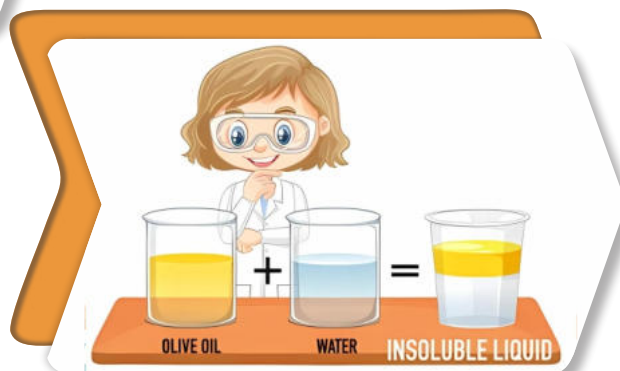
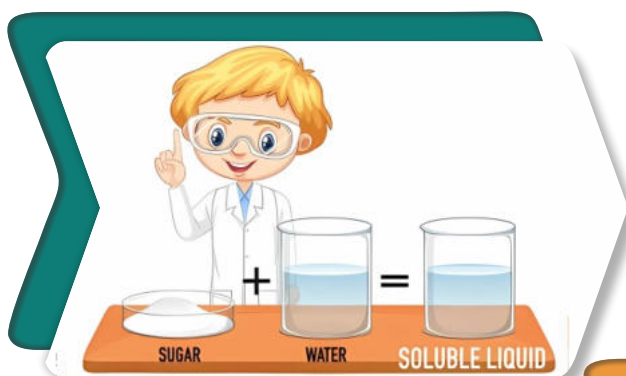
Mixtures are physical combinations of two or more substances where each substance retains its properties.

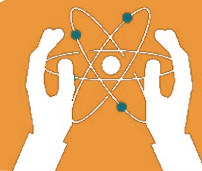


Types:

1.Homogeneous Mixture: A mixture that has the same composition throughout (e.g., sugar and water solution).

2.Heterogeneous Mixture: A mixture with visibly different parts (e.g., water and olive oil).





ACTIVITY

Materials: Sand, water, salt, oil, beakers.

Instructions:

Mix sand with water in one beaker, salt with water in another and oil with water into next beaker. Observe each mixture to identify whether it is homogeneous or heterogeneous.

Methods of Separating Mixtures

A. Filtration (For Insoluble Solids in Liquids)

Filtration is used to separate insoluble solids from a liquid using a filter.

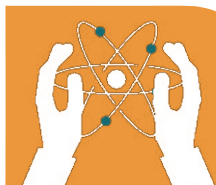
Experiment:

- Take a mixture of water and sand.
- Place a filter paper (or a fine cloth) in a funnel.
- Pour the mixture through the funnel.
- Observe how the sand remains in the filter, while clean water passes through.

Questions:

1. What happens to the sand in this experiment?

2. Can filtration be used to separate salt from water? Why or why not?



B. Evaporation:

Evaporation is used to separate a dissolved solid from a liquid by heating.

Experiment:

- Take a saltwater solution in a small dish.
- Heat the dish over a flame or under sunlight.
- Observe how the water evaporates and salt is left behind.

Questions:

1. What happens to the water during evaporation?

2. Why does salt remain in the dish after heating?

C. Magnetic Separation (For Magnetic and Non-Magnetic Substances)

Used for separating magnetic materials (like iron) from non-magnetic ones.

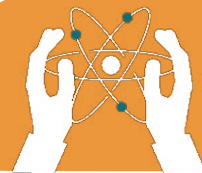
Experiment:

- Take a mixture of iron filings and sand.
- Use a magnet to attract the iron filings.
- Observe how the iron filings stick to the magnet, leaving the sand behind.

Questions:

1. Why does the magnet attract only the iron filings?

2. Can this method be used to separate plastic from sand? Why or why not?



D. Distillation:

Distillation is a physical separation technique used to separate a liquid from a mixture based on differences in boiling points. It involves heating the liquid to produce vapour and then cooling the vapour to collect the purified liquid.

Experiment:

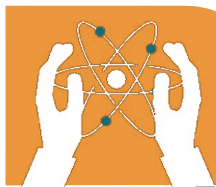
1. pour the saltwater solution into a conical flask.
2. Attach a rubber stopper with a delivery tube and place the other end in a beaker.
3. Heat the flask gently.
4. Observe as water evaporates, moves through the tube, and condenses in the beaker.
5. Compare the collected water with the original saltwater solution.

Questions:

1. What remains in the conical flask after heating?

2. Why does only water evaporate and not the salt?

3. How can distillation be used in daily life?

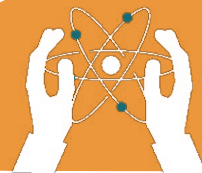


ELEMENTS, COMPOUNDS, AND MIXTURES

Worksheet:

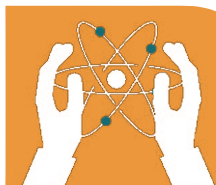
Identify each diagram as either an element, compound, or mixture.

Air	Oxygen	Water
1.	2.	3.
Carbon Dioxide	Nitrogen	Helium
4.	5.	6.



Fill in the blanks.

1. An element is a pure substance made up of only one type of _____.
2. A compound is a substance formed when two or more elements are _____ bonded.
3. _____ mixtures have the same composition throughout, while _____ mixtures have visibly different parts.
4. Distillation separates substances based on their _____ points.
5. _____ is a process used to separate solids from liquids.



ELEMENTS, COMPOUNDS, AND MIXTURES

Mark Tick ✓ or Cross ✗ in the following statements.

I. A compound has properties that are different from the elements that form it.

☐☐

II. Air is an example of a homogeneous mixture.

☐☐

III. Elements can be broken down into simpler substances by chemical methods.

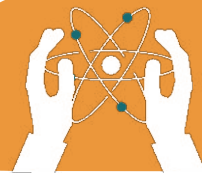
☐☐

IV. A mixture can be separated into its components by physical methods.

☐☐

V. In a compound, the elements are chemically bonded in a fixed ratio.

☐☐



Answer the Following Questions:

Short-Answer Questions

- Q1. What is an element? Give two examples.
Q2. Define a compound and provide two examples.
Q3. What is a mixture? Define its types.
Q4. Define metals, non-metals, and metalloids.

Long-Answer Questions

- Q1. Describe the differences between elements, compounds, and mixtures in terms of composition, properties, and separation methods, with examples.



Critical Thinking

If you are given an unknown substance, how would you determine whether it is an element or a compound?